



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Fall-24_25

Section: A

Software Quality Assurance and Testing

Rent A House Platform

A Report submitted
By

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Date:

Software Test Plan

For

Rent A House Platform

Version 0.6 approved

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American International University of Bangladesh

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Table of Contents

Revision History	3
1. TEST PLAN IDENTIFIER: RS-MTP01.3	4
2. REFERENCES	4
3. INTRODUCTION	4
3.1 Background to the Problem:	4
4. REQUIREMENT SPECIFICATION	5
4.1 System Feature	5
4.2 System Quality Attributes	8
4.3 System Interface.....	9
4.4 Project Requirements	18
5. FEATURES NOT TO BE TESTED.....	19
6. TESTING APPROACH	19
6.2 Test Tools	20
6.2.1 Bug Tracking:.....	20
6.2.2 Test Script Writing and Execution:	20
6.3 Meetings	20
7. TEST CASES/TEST ITEMS	21
8. ITEM PASS/FAIL CRITERIA.....	38
9. TEST DELIVERABLES	38
10. STAFFING AND TRAINING NEEDS.....	39
11. RESPONSIBILITIES	39
12. TESTING SCHEDULE	40
13. PLANNING RISKS AND CONTINGENCIES	41
14. APPROBALS.....	41

Revision History

Revision	Date	Updated by	Update Comments
0.1	2025.01.19		Initial Document
0.2	2025.01.22		First Draft
0.3	2025.01.24		Second Draft
0.4	2025.01.26		Updated Requirement
0.5	2025.01.27		Updated Test Cases
0.6	2025.02.01		Final Revision

1.TEST PLAN IDENTIFIER: RS-MTP01.3

2. REFERENCES

- ✓ Software Quality and Testing Course PowerPoint Slides
- ✓ Software Requirement documentation
- ✓ Software Requirements Specification (SRS) Document

3. INTRODUCTION

3.1 Background to the Problem:

The housing rental market is witnessing a significant increase in demand, highlighting the need for a transparent, efficient, and user-friendly platform to connect property owners and tenants. Traditional methods of renting properties often lack accessibility, efficiency, and transparency, making the process cumbersome for both parties. While existing platforms like Zillow, Airbnb, Rent.com, and bproperty have demonstrated the potential of digital solutions, challenges remain, such as limited user engagement, inefficiencies in communication, and lack of localized features. Addressing these gaps is crucial to creating a streamlined and inclusive rental experience.

3.2 Solution to the Problem:

The "Rent A House" project proposes a comprehensive web-based platform that simplifies the rental process and bridges the gap between property owners and tenants. This platform offers user-friendly property listings, robust search capabilities, and secure booking systems to enhance accessibility and trust. Key features include detailed property information for owners, efficient search and discovery for tenants, and secure transaction processing through an integrated payment gateway. Additionally, user profiles, review systems, and real-time notifications foster transparency and engagement, positioning "Rent A House" as a reliable and innovative solution for transforming the real estate rental market.

4. REQUIREMENT SPECIFICATION

4.1 System Features

4.1.1 System Signup

Functional Requirements:

- i. The system shall allow new users to sign up by providing their email, username, password, and any required personal details.
- ii. The password must meet security requirements, such as a minimum length, inclusion of uppercase letters, numbers, and special characters.
- iii. If any required fields are incomplete or invalid, the system shall display an appropriate error message and prevent the signup from proceeding.
- iv. The system shall not allow duplicate email addresses or usernames during signup.

Priority Level: High

Precondition: The user must provide valid and unique information for all required fields.

4.1.2 System Login

Functional Requirements:

- i. The system shall allow users to log in with their given username and password.
- ii. If the username and/or password is entered incorrectly more than three times, the system shall generate a random verification code to retry login.
- iii. If the number of login attempts exceeds the limit (5 times), the system shall block the user's account for one hour.

Priority Level: High

Precondition: The user must have a valid username and password.

4.1.3 Property Listing

Functional Requirements:

- i. The system shall allow property owners to input detailed information about their properties, including descriptions, location, and features.
- ii. The system shall allow property owners to upload images to showcase the property visually.
- iii. The system should enable property owners to manage the pricing and availability of their properties.

Priority Level: High

Precondition: The user must be a registered property owner.

4.1.4 Search and Filtering

Functional Requirements:

- i. The system should provide advanced search options.
- ii. Allowing users to filter properties based on location, price range, and specific features.

Priority Level: High

Precondition: The user must have a valid account.

4.1.5 Booking System

Functional Requirements:

- i. The system shall provide a secure messaging system for communication between tenants and property owners.
- ii. The system shall allow tenants to send reservation requests to property owners.
- iii. The system shall enable property owners to confirm or reject reservation requests.
- iv. The system shall include calendar integration to display property availability.

Priority Level: High

Precondition: The user must have a verified account and an active booking request.

4.1.6 User Profiles and Reviews

Functional Requirements:

- i. The system shall allow property owners and tenants to create and manage profiles.
- ii. The system shall include a review and rating system to foster trust between users.
- iii. The system shall display aggregated reviews and ratings on user profiles.

Priority Level: Medium

Precondition: The user must have a valid account.

4.1.7 Notification System

Functional Requirements:

- i. The system shall send real-time notifications to users for booking requests, confirmations, and updates.
- ii. The system shall notify users about any changes to property availability or pricing.
- iii. The system should send reminders for upcoming booking events or payments.

Priority Level: Medium

Precondition: The user must have an active account and permissions for notifications.

4.1.8 Messaging with Owner

Functional Requirements:

- i.** The system shall allow tenants to send direct messages to property owners for inquiries about the property.
- ii.** The system shall notify property owners of new messages from tenants.
- iii.** The system shall provide a messaging interface where both tenants and property owners can view their message history.
- iv.** The system shall restrict messaging to verified users only to prevent spam.

Priority Level: High

Precondition: Both tenants and property owners must have verified accounts

4.1.9 Payment System

Functional Requirements:

- i.** The system shall provide a secure payment gateway for processing transactions.
- ii.** The system allows users to select various payment methods (e.g., credit card, debit card, bank transfer).
- iii.** The system shall send a payment confirmation notification to users after successful transactions.
- iv.** The system shall display a detailed receipt for each payment in the user's payment history.
- v.** The system will update the booking status to "Paid" after successful payment.

Priority Level: High

Precondition: The user must be logged in, have a confirmed booking, and access to a valid payment method.

4.2 System Quality Attributes

1. **Usability:**
The system should allow users to search for houses, communicate with property owners, and make bookings effortlessly, ensuring a smooth, user-friendly experience.
2. **Reliability:**
The system should operate consistently without unexpected downtime, ensuring users can search for properties, send messages, and book houses reliably.
3. **Performance:**
The system should respond promptly to user actions, with fast search results, seamless map loading, and quick property listing updates to provide an efficient experience.
4. **Security:**
The system should protect user data, including personal information, payment details, and messaging history, ensuring confidentiality and safeguarding against unauthorized access or breaches.
5. **Testability:**
The system should be designed to facilitate all levels of testing (unit, integration, and system testing), ensuring validation and verification of key features like property search, booking, and messaging.
6. **Maintainability:**
The system should be built using modular components and have clear documentation, enabling easy updates, bug fixes, and enhancements to functionality.
7. **Interoperability:**
The system should integrate seamlessly with third-party services such as payment gateways, mapping APIs (like Google Maps), and notification systems (email or SMS).
8. **Accessibility:**
The system should comply with accessibility standards, ensuring it can be used effectively by users of all abilities, including those with disabilities.
9. **Robustness:**
The system should handle unexpected failures gracefully, such as failed payments or server crashes, ensuring the user experience is not significantly disrupted.
10. **Efficiency:**
The system should utilize resources efficiently, ensuring optimal performance while minimizing loading times and reducing resource consumption, providing a cost-effective and smooth experience for all users.

4.3 System Interface



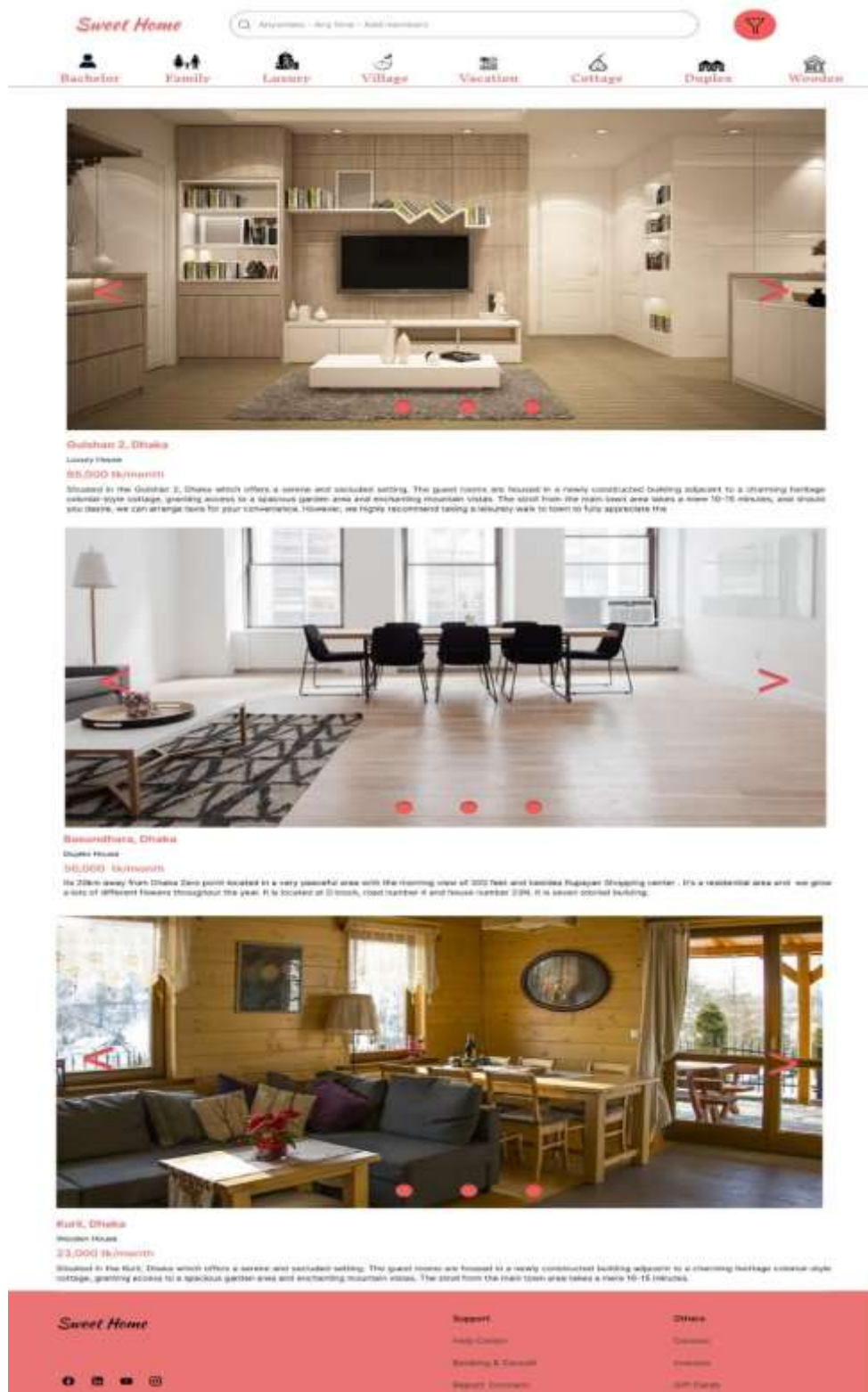


Fig1: Home page



\$60 night

CHECK-IN 4/1/2024	CHECKOUT 4/5/2024
GUESTS 1 guest	

[Reserve](#)

Gulshan 2, Dhaka

Luxury House

\$650 month

Situated in the Gulshan 2, Dhaka which offers a serene and secluded setting. The guest rooms are housed in a newly constructed building adjacent to a charming heritage colonial-style cottage, granting access to a spacious garden area and enchanting mountain vistas. The stroll from the main town area takes a mere 10-15 minutes, and should you desire, we can arrange taxis for your convenience.

What this place offers

- Lock on bedroom door
- Wifi
- TV
- Private patio or balcony
- Bathtub
- Smoke alarm
- Free parking on premises
- Dedicated workspace
- Well Furnished
- Garden view
- Security cameras on property
- refrigerator
- Washing Machine
- Hair dryer
- Hot water
- Essentials

Cancellation policy

Before you book, make sure you're comfortable with this Host's cancellation policy. Keep in mind that Airbnb's [Extenuating Circumstances policy](#) doesn't cover cancellations due to illness

Details Picture



★★★★★

Had a great time in Gulshan 2 thanks to this lovely stay. The rooms were better than the given pics. Very responsive manager in Pradip. I love the place



Miliy

Dhaka, Bangladesh

★★★★★

the place to be. If you want to alone or as a couple. Mary is very very friendly and made us feel at home. not to miss her mom's food. made us feel good



Rakib

Comilla, Bangladesh

★★★★★

The stay was amazing, Mary was very helpful. She went out of way to arrange our transport right from the airport. We felt at home. Everything was excellent



Sultan

Rangpur, Bangladesh

★★★★★

Was there for two days, from sunrise till sunset had a great peaceful time! Interiors are amazing! Colour was so good for coordination of the room



Zerin

Rajshahi, Bangladesh

Fig2: Property Details

×

Filters

Type of Place

Any Type

Room

House

Price Range

Min

—

Max

Bedrooms

Any

1

2

3

Bathrooms

Any

1

2

3

Property Type



Bachelor



Family



Cottage



Luxury

Essentials

Wifi

☐

Kitchen

☐

TV

☐

Hot water

☐

Show more

Booking Options

Instant Book

☒

Self Check-in

☐

Allow Pets

☐

Fig3: Advance Filtering



Sing Up



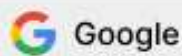
Sweet Home

Create an account

Enter your email to sign up for this app

Sign up with email

or continue with



Google

By clicking continue, you agree to our [Terms of Service](#)
and [Privacy Policy](#)

Fig4: Sign-Up

Review

Write your reviews

Submit

Fig5: Review

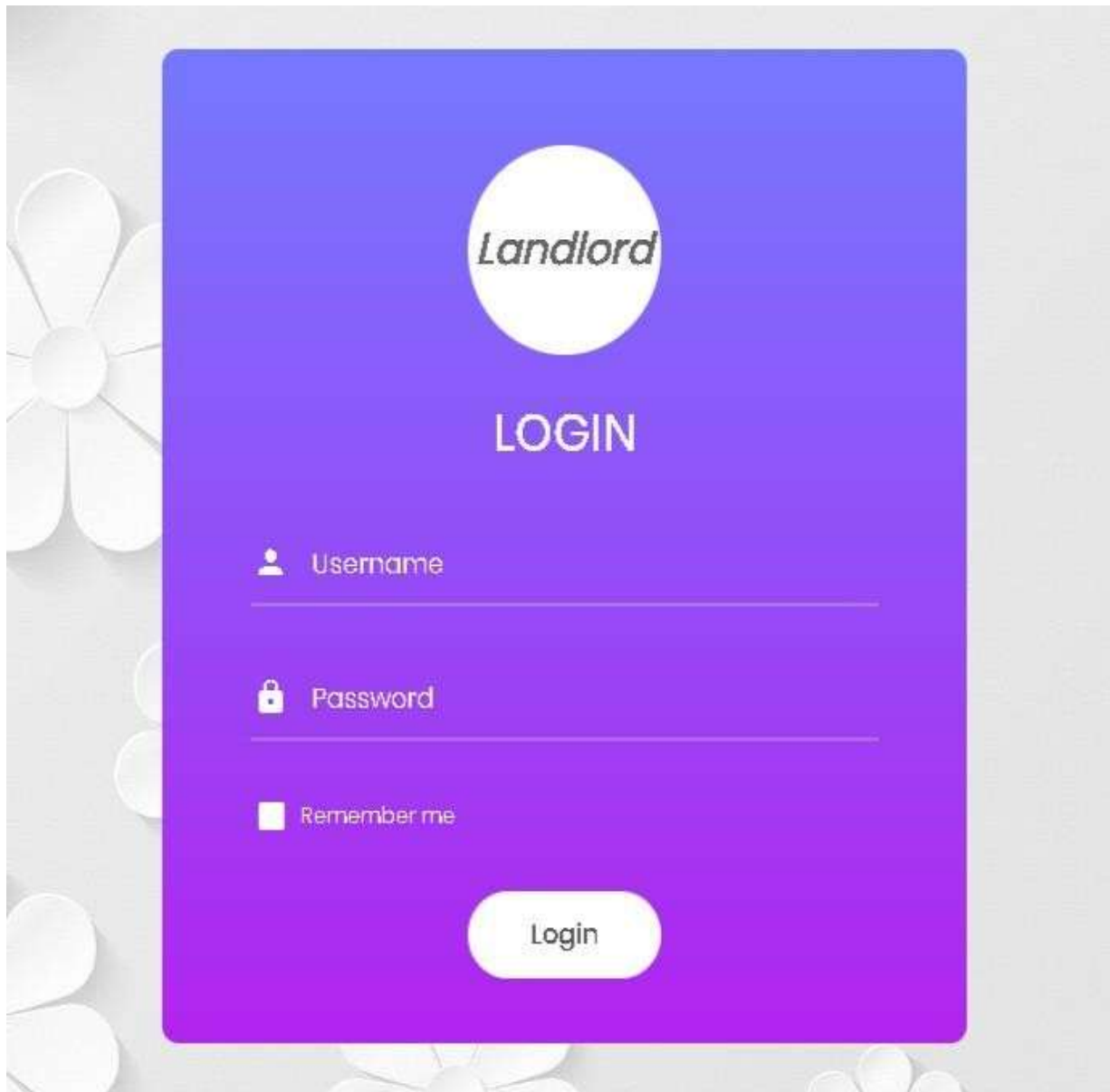


Fig6: Sign-In

Property

Upload Pictures



Property Name

Victory House

Price per Night/Month

20,000 tk/Month, 1500 tk/Night

Property Address

Kuratoli, Dhaka, Bangladesh

Google Map Link

<https://maps.app.goo.gl/YMCCnEspeDpEHDKCA>

Property Type

Bachelor

Number of Bedrooms

4

Number of Washroom

3

Essential

TV

Wifi

Hot Water

+ Add another

Description

Upload Videos



Save changes

Fig7: Add Property Details

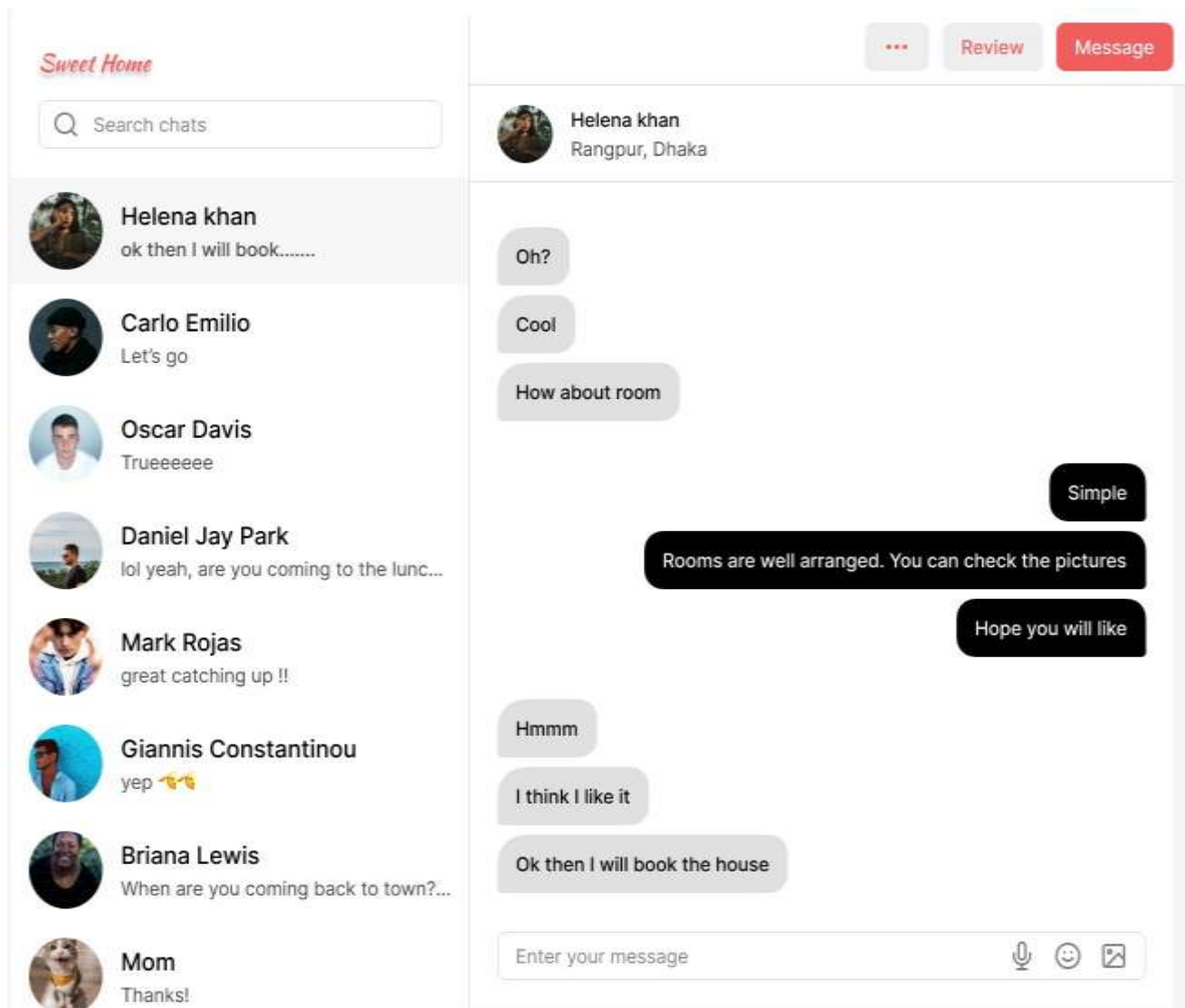


Fig8: Messaging

4.4 Project Requirements

Budget estimation:

Resource Name	Type	Cost(TAKA)
Project Management Team	Work	20,000
HR Team	Work	20,000
System Designer Team	Work	10,000
Software Development Team	Work	10,000
Quality Assurance Team	Work	10,000
System Testing Team	Work	10,000
Number of Computer	Recourse	150,000
Server	Resource	40,000
Others	Recourse	30,000
Total		3,00,000

Time estimation:

Task Name	Duration
Documentation	7 days
Design	10 days
Test plan	7 days
Unit testing	3 days
End-End to Testing	10 days
Bug Reporting and Fixing	5 days
System Testing	5 days
System testing bug report & bug fixing	3 days
Acceptance testing	3 days
Acceptance test bug report & bug fixing	3 days
Feedback & Modification	4 days
Total	60 days

5. FEATURES NOT TO BE TESTED

The following is a list of the areas that will not be specifically addressed for testing in this project. All testing in these areas will be indirect because of other testing efforts:

- **Performance Testing of Third-Party Systems:** As the project primarily focuses on internal software functionalities, the performance of external third-party systems will not be directly tested. Any performance issues related to these systems will be monitored but not formally tested.
- **Load Testing:** Load testing scenarios involving a significant number of concurrent users accessing the system simultaneously will not be conducted as part of this testing phase. This includes stress testing to evaluate system behavior under extreme user loads.

6. TESTING APPROACH

6.1 Testing Levels

For the **Rent a House** project, the testing strategy will consist of three key levels: Unit Testing, System/Integration Testing, and Acceptance Testing.

Unit Testing : Unit Testing will be conducted by the developers, who will be responsible for testing individual components of the system, such as user authentication (login/signup), property listing and image uploads, search and filter functionalities, and messaging system operations. Before the unit tests are accepted, developers must provide comprehensive evidence, including test case lists, sample outputs, data printouts, and defect reports. This information will be reviewed by the development team leader and shared with the testing team for validation.

System/Integration Testing: System/Integration Testing will involve collaboration between the test manager and the development team leader, with support from individual developers as required. This phase will focus on validating the integration of all system modules to ensure that functionalities such as property search, booking, notifications, and messaging work seamlessly together. Programs will only transition to this phase once all critical defects have been resolved. While up to two major defects may be present, they must not block the testing process, and acceptable workarounds should be available. Testing will primarily be manual, as no specific testing tools are designated for this project.

Acceptance Testing: Acceptance Testing will involve actual end users, including property owners and tenants, who will assess the system's ability to meet both functional and non-functional requirements such as usability, reliability, and security. This phase will run for one month, concurrently with the existing manual processes, following the completion of System/Integration Testing. The focus will be on verifying accurate property listings, effective booking workflows, secure communication between tenants and owners, reliable notification delivery, and overall user satisfaction in real-world conditions.

6.2 Test Tools

6.2.1 Bug Tracking:

- Tool: JIRA
- Purpose: Used to track and manage reported bugs, issues, and defects throughout the testing process. It will allow the team to assign, prioritize, and monitor the progress of bug fixes.

6.2.2 Test Script Writing and Execution:

- Tool: Qase
- Purpose: Qase will be used to write, manage, and execute test scripts for various testing phases. It will help in creating structured test cases for Unit Testing, System/Integration Testing, and Acceptance Testing. The results from test executions will be tracked and analyzed within Qase to ensure all test scenarios are covered.

6.3 Meetings

The test team will meet once every week to evaluate progress, review test results, and identify any error trends or issues as early as possible. The test team leader will meet with the development team and the project manager once every two weeks to discuss overall testing progress, critical defects, and necessary improvements. These meetings will be scheduled in alternating weeks to ensure continuous monitoring and collaboration.

Additionally, urgent meetings may be scheduled as required in case of critical system failures, unexpected defects, or major changes to project requirements. The goal of these meetings is to ensure smooth testing execution, address any bottlenecks, and facilitate a seamless transition from development to deployment.

7. TEST CASES/TEST ITEMS

Project Name: Sign Up		Test Designed by: Noman Islam		
Test Case ID: SU_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Nasimul Islam		
Module Name: Sign up session		Test Execution date: 16-april- 24		
Test Title: Verify sign up process				
Description: Test the sign-up process to ensure users can successfully create an account on the RentA-House platform.				
Precondition (If any): User must have a valid email account.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Rent-A House platform. 2. Locate the sign-up page. 3. Fill in the sign-up form with valid user details (e.g.,username, email, password). 4. Click on the sign-up button. 5. Verify if the user is redirected to the login page after successful sign up.	Username: minahj42 Email:minhaj42@gmail.com Password: at@45.# Confirm Password: at@45.#	User should be successfully signed up and redirected to the login page.	As expected	Pass
Post Condition: User account is created in the system, and the user is redirected to the login page.				

Project Name: Sign Up		Test Designed by: Noman Islam		
Test Case ID: SU_2		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Nasimul Islam		
Module Name: Sign up session		Test Execution date: 16-april- 24		
Test Title: Verify sign-up process for an existing email address (Duplicate Email)				
Description: Test the sign-up process to ensure that the system prevents users from signing up with an email that is already registered.				
Precondition (If any): The email address must already be registered in the system.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/F ail)
1. Navigate to the Rent-A House platform. 2. Locate the sign-up page. 3. Fill in the sign-up form with a duplicate email address (one that is already registered) and valid user details (e.g., username, email, password). 4. Click on the sign-up button.	Username: johndoe22 Email: minhaj42@gmail.com (Already registered) Password: at@45.# Confirm Password: at@45.#	The system should display an error message stating that the email is already in use and prevent further registration.	As expected	Pass
Post Condition: No new account is created, and the user remains on the sign-up page with an error message displayed.				

Project Name: Sign Up		Test Designed by: Noman Islam		
Test Case ID: SU_3		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Nasimul Islam		
Module Name: Sign up session		Test Execution date: 16-april- 24		
Test Title: Verify sign-up process for incomplete or invalid fields				
Description: Test the sign-up process to ensure that the system displays error messages and prevents the user from proceeding if any required fields are incomplete or invalid.				
Precondition (If any): User must access the sign-up form.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Rent-A House platform. 2. Locate the sign-up page. 3. Fill in the sign-up form with incomplete or invalid data in the required fields. 4. Attempt to submit the form with an invalid email format. 5. Leave the "Confirm Password" field blank.	Username: johndoe22 Email: johndoe@gmail.com Password: at@45. Confirm Password:	The system should display an error message: "Confirm Password is required."	As expected	Pass
Post Condition: User is unable to complete the sign-up process if any required fields are missing or invalid. The system shows specific error messages for each invalid field.				

Project Name: Login		Test Designed by: Noman Islam		
Test Case ID: LG_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Nasimul Islam		
Module Name: Login Session		Test Execution date: 16-april-24		
Test Title: Verify the lock out for multiple unsuccessful login attempt				
Description: Test the login functionality to ensure users can successfully log in to the Rent-AHouse platform.				
Precondition (If any): User account must be registered.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Locate the login page. 3. Enter valid credentials (username/email and password). 4. Click on the login button. 5. Verify if the user is redirected to the home page after successful login.	Username: minhaj45 Password: at@45.#	User should be successfully logged in and redirected to the home page.	As expected	Pass
Post Condition: User session is established, and the user is redirected to the home page.				

Project Name: Login		Test Designed by: Noman Islam		
Test Case ID: LG_2		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Nasimul Islam		
Module Name: Login Session		Test Execution date: 16-april-24		
Test Title: Verify generation of random verification code after multiple failed login attempts.				
Description: Test the login functionality to ensure that after more than three unsuccessful login attempts, the system generates a random verification code to retry login.				
Precondition (If any): User account must be registered.				
Test Steps	Test Data	Expected Results	Actual Result	Status (Pass/Fail)
1. Navigate to the Rent-A House platform. 2. Locate the login page. 3. Enter an incorrect username or password and click on the login button. 4. Repeat the above step for 2 more attempts with incorrect credentials. 5. After the third failed attempt, the system should display a message prompting for a verification code. 6. Enter the verification code and attempt to log in again.	Verification Code: (Code shown by system) Username: johndoe45 Password: correctpassword	The system should allow the user to log in successfully if the verification code is entered correctly.	As expected	Pass
Post Condition: The system should allow login only after the correct verification code is provided.				

Project Name: Login		Test Designed by: Noman Islam		
Test Case ID: LG_3		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Nasimul Islam		
Module Name: Login Session		Test Execution date: 16-april-24		
Test Title: Verify account lockout after exceeding the login attempt limit.				
Description: Test the login functionality to ensure that after exceeding the maximum number of allowed login attempts (5 times), the system locks the user's account for one hour.				
Precondition (If any): User account must be registered.				
Test Steps	Test Data	Expected Results	Actual Result	Status (Pass/Fail)
1. Navigate to the Rent-A House platform. 2. Locate the login page. 3. Enter an incorrect username or password and click on the login button (first attempt). 4. Repeat the above step for 4 more attempts (total 5 failed attempts). 5. After the 5th failed attempt, verify if the user is locked out and an appropriate error message is displayed. 6. Attempt to log in again within the one-hour lockout period. 7.Wait for one hour and then attempt to log in again.	Username: johndoe45 Password: wrongpassword (1 st ,2 nd ,3 rd ,4 th ,5 th attempts)	The system should lock the account and display a message like "Your account has been locked for one hour due to multiple failed login attempts."	As expected	Pass
	Username: johndoe45 Password: correctpassword	After one hour, the user should be able to log in successfully using the correct credentials.	As expected	Pass
• Post Condition: After one hour, the user should be able to log in again without any issues.				

Project Name: Logout		Test Designed by: Noman Islam		
Test Case ID: LG_2		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Khadiza Akter		
Module Name: Logout Session		Test Execution date: 16-april-24		
Test Title: Verify logout functionality				
Description: Test the logout functionality to ensure users can successfully log out of the Rent-AHouse platform.				
Precondition (If any): User must be logged in.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Locate the logout option in the user profile or settings. 3. Click on the logout option. 4. Verify if the user is redirected to the login page after successful logout.	N/A	User should be successfully logged out and redirected to the login page.	As expected	Pass
Post Condition: User session is terminated, and the user is redirected to the login page.				

Project Name: Add a property		Test Designed by: Noman Islam		
Test Case ID: AP_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Minhajul Islam		
Module Name: Add a Property Session		Test Execution date: 16-april-24		
Test Title: Verify property addition process				
Description: Test the process of adding a property to the Rent-A-House platform.				
Precondition (If any): User must be logged in as a property owner.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Log in to the platform as a property owner with valid credentials. 3. Find the option to add a new property. 4. Fill in property details (e.g., property type, location, amenities, price). 5. Upload property images. Submit the property listing. 6. Verify if the property is successfully added to the platform.	Property Type: Bachelor Location: Khilkhet, Dhaka Amenities: Hot water, Parking area, Security guard, Wifi Price: 25000/month No. of Room: 4 No. of Bathroom: 3 Map Link: https://maps.app.google.com/YUSB	Property should be successfully added to the platform, and the user should receive a confirmation message.	As expected	Pass
Post Condition: Property listing is recorded in the system, and the user receives confirmation.				

Project Name: Remove a Property		Test Designed by: Noman Islam		
Test Case ID: RP_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Noman Islam		
Module Name: Remove a Property Session		Test Execution date: 16-april-24		
Test Title: Verify property removal process				
Description: Test the process of removing a property from the Rent-A-House platform.				
Precondition (If any): Property to be removed must exist on the platform and belong to the logged-in owner.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Log in to the platform as a property owner with valid credentials. 3. Locate the "My Properties" or "Manage Properties" section. 4. Select the property to be removed. 5. Click the "Remove" or "Delete" button and confirm the removal action. 6. Verify if the property is no longer listed on the platform.	Username: owner123 Password: owner@45 Property ID: 1001 Confirmation Action: "Yes"	Users should see a list of their added properties. The system should display the selected property's details and a "Remove" option. The system should prompt the user to confirm the removal. Upon confirmation, the property should be removed from the platform.	As expected	Pass
Post Condition: The property is successfully removed from the system.				

Project Name: Edit Property		Test Designed by: Noman Islam		
Test Case ID: EP_1		Test Designed date: 19-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Nasimul Islam		
Module Name: Edit Property Session		Test Execution date: 18-april-24		
Test Title: Verify property editing functionality				
Description: Test the functionality to edit a property listing on the Rent-A-House platform.				
Precondition (If any): User must be logged in as the property owner.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Log in to the platform as the property owner with valid credentials. 3. Locate the property to be edited in the property management section. 4. Click on the property to access the edit options. 5. Modify the property details (e.g., update location, amenities, price). 6. Save the changes. Verify if the changes are successfully updated.	Property ID:24 Updated Location: Uttara, Dhaka Updated Amenities: Hot water, Parking area, Security guard, Wi-Fi, Washing machine Updated Price: 30000/-	Property details should be successfully updated, and the user should receive a confirmation message.	As expected	Pass
Post Condition: Property listing is updated in the system, and the user receives confirmation.				

Project Name: Reserve a Room		Test Designed by: Noman Islam		
Test Case ID: RR_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Noman Islam		
Module Name: Reserve a Room Session		Test Execution date: 16-april-24		
Test Title: Verify room reservation process				
Description: Test the process of reserving a room on the Rent-A-House platform.				
Precondition (If any): User must be logged in.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Log in to the platform with valid credentials. 3. Search for available rooms based on desired criteria (e.g., location, category, price). 4. Select a room from the search results. 5. Choose reservation details (e.g., check-in date, check-out date, number of guests). 6. Proceed to reserve the selected room. 7. Verify if the reservation is confirmed.	Check-in Date: 17/04/24 Check-out Date: 20/04/24 Number of Guests: 4	Room reservation should be successfully confirmed, and the user should receive a confirmation message or email.	As expected	Pass
Post Condition: Room reservation is recorded in the system, and the user receives confirmation.				

Project Name: Cancel Reservation		Test Designed by: Noman Islam		
Test Case ID: CR_1		Test Designed date: 19-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Khadiza Akter		
Module Name: Cancel Reservation Session		Test Execution date: 18-april-24		
Test Title: Verify reservation cancellation process				
Description: Test the process of canceling a reservation on the Rent-A-House platform..				
Precondition (If any): User must be logged in and have a confirmed reservation.				
Test Steps	Test Data	Expected Results	Actual Result	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Log in to the platform with valid credentials. 3. Go to the reservation details page. 4. Locate the option to cancel the reservation. 5. Confirm the cancellation. 6. Verify if the reservation is successfully canceled.	Cancel Reservation ID : 24	Reservation should be successfully canceled, and the user should receive a confirmation message.	As expected	Pass
Post Condition: Reservation should be successfully canceled, and the user should receive a confirmation message.				

Project Name: Search and Filtering		Test Designed by: Noman Islam		
Test Case ID: SF_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Noman Islam		
Module Name: Search and Filtering		Test Execution date: 16-april-24		
Test Title: Verify search and filtering functionality for property listings.				
Description: Test the advanced search and filtering options on the Rent-A-House platform to ensure users can find properties based on location, price range, and specific features.				
Precondition (If any): Property listings must exist in the system.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Locate the search bar on the homepage. 3. Enter a keyword or location in the search bar and click "Search." 4. Apply a price range filter. 5. Apply a specific feature filter (e.g., number of bedrooms). 6. Apply multiple filters (e.g., location, price, and features) simultaneously. 7. Clear all filters and verify the reset functionality.	Location: Dhaka	The system should display relevant properties located in Dhaka.	As expected	Pass
	Min Price: 10,000 BDT Max Price: 20,000 BDT	The system should filter properties to display only those within the specified price range.	As expected	Pass
	Feature: 2 Bedrooms	The system should display properties that match all the selected criteria.	As expected	Pass
	Location: Dhaka Price: 10,000–20,000 BDT Feature: 2 Bedrooms	The system should display all available properties without any filters applied.	As expected	Pass
Post Condition: ☐ Users can successfully filter properties by location, price range, and specific features. ☐ Search results dynamically update based on the filters applied. ☐ Users can reset filters to view all properties.				

Project Name: Messaging System		Test Designed by: Noman Islam		
Test Case ID: MSG_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Nasimul Islam		
Module Name: Messaging		Test Execution date: 16-april- 24		
Test Title: Verify messaging functionality between tenants and property owners.				
Description: Test the messaging system to ensure tenants and property owners can exchange messages and view message history.				
Precondition (If any): Both tenant and property owner accounts must be verified.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Navigate to the Rent-A House platform. 2. Log in to the system as a tenant. 3. Locate a property and send an inquiry message to the property owner. 4. Log in as the property owner and respond to the message. 5. Verify the messaging interface for both users.	Username: tenant123 Password: tenant@123 Message Content: "Is the property available for the selected date?" Response Content: "Yes, it is available."	The property owner receives a real-time notification about the message. The tenant receives a notification about the reply, and the message history is updated. The message history shows a complete log of the conversation.	As expected	Pass
Post Condition: Both users can view the message history in the messaging interface.				

Project Name: Notification System		Test Designed by: Noman Islam		
Test Case ID: NS_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Nasimul Islam		
Module Name: Notifications		Test Execution date: 16-april- 24		
Test Title: Verify real-time notifications for booking updates				
Description: Test if the system sends real-time notifications for booking requests, confirmations, and updates.				
Precondition (If any): The user must have an active account and enabled notifications.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/F ail)
1. Navigate to the Rent-A House platform. 2. Log in to the system as a tenant. 3. Initiate a booking request for a property. 4. Confirm the booking as the property owner. 5. Modify the property pricing or availability.	Username: tenant123 Password: tenant@123 Property ID: 2001 Booking Date: 20/04/24 Confirmation Action: "Approve" Price updated to 28,000/month	The system sends a booking confirmation notification to the tenant in real time.	As expected	Pass
Post Condition: Notifications are successfully received in real time for booking requests, confirmations, and updates.				

Project Name: Submit Review		Test Designed by: Noman Islam		
Test Case ID: SR_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Noman Islam		
Module Name: Submit Review Session		Test Execution date: 16-april-24		
Test Title: Verify review submission process				
Description: Test the process of submitting a review for a property on the Rent-A-House platform.				
Precondition (If any): User must be logged in and have booked and stayed at the property.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Log in to the platform with valid credentials. 3. Click on the property to view details. 4. Find the option to submit a review. 5. Enter review details (e.g., rating, comments). 6. Submit the review. 6. Verify if the review is successfully submitted.	Rating: 5 Comments: Environment of the house is good. The description is valid. The Owner is a cool person.	Review should be successfully submitted, and the user should receive a confirmation message.	As expected	Pass
Post Condition: Review is recorded for the property in the system, and the user receives confirmation.				

Project Name: Make Payment		Test Designed by: Noman Islam		
Test Case ID: MP_1		Test Designed date: 17-april-24		
Test Priority (Low, Medium, High): HIGH		Test Executed by: Minhajul Islam		
Module Name: Make Payment Session		Test Execution date: 16-april-24		
Test Title: Verify payment process				
Description: Test the process of making a payment for a booking on the Rent-A-House platform.				
Precondition (If any): User must be logged in and have a confirmed booking.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Rent-AHouse platform. 2. Log in to the platform with valid credentials. 3. Go to the booking details page. 4. Locate the option to make a payment. 5. Select a payment method (e.g., credit card). 6. Enter payment details (e.g., card number, expiration date, CVV). 7. Confirm the payment. 8. Verify if the payment is successful.	Payment Method: Card Card Number: 471862*** Expiration Date: 08/26 CVV: ***	Payment should be successfully processed, and the user should receive a confirmation message or email.	As expected	Pass
Post Condition: Payment is recorded in the system, and the booking status is updated to "Paid".				

8. ITEM PASS/FAIL CRITERIA

The completion of the test process for the **Rent A House** platform depends on the successful execution of several critical functionalities. Firstly, registered users should be able to search for properties, book them, and communicate with property owners without encountering any technical issues or errors. When a user sends a booking request and the property owner confirms it, the system should accurately update the property's availability status and notify both parties.

The messaging system between tenants and property owners must function seamlessly, ensuring secure and reliable communication. Notifications for booking requests, confirmations, and updates should be delivered promptly and without any delays. Any discrepancies, such as failed notifications, errors in booking processes, or issues in property availability updates, will signify a failure in the test criteria.

Conversely, if users can effortlessly search for properties, book them, communicate with owners, and receive notifications as expected, it indicates that the platform has met the test criteria successfully for the **Rent A House** project.

9. TEST DELIVERABLES

9.1 Test Plan

A detailed document outlining the testing approach, scope, objectives, resources, schedule, and risks for all phases of testing, including Unit Testing, System/Integration Testing, and Acceptance Testing.

9.2 Traceability Matrix

The Traceability Matrix is a key deliverable in the testing process that ensures every requirement specified in the project has been addressed through testing. It links functional requirements to their respective test cases and helps track the testing coverage and status.

9.3 Test Summary

A report summarizing the results of the testing phases, including metrics such as test cases executed, passed, failed, defects found, and their resolution status. It will also include a final recommendation for the system's readiness for deployment.

10. STAFFING AND TRAINING NEEDS

At least one (1) full-time tester should be assigned for system/integration and acceptance testing, transitioning from a part-time role after four months. If unavailable, the project/test manager will assume this role.

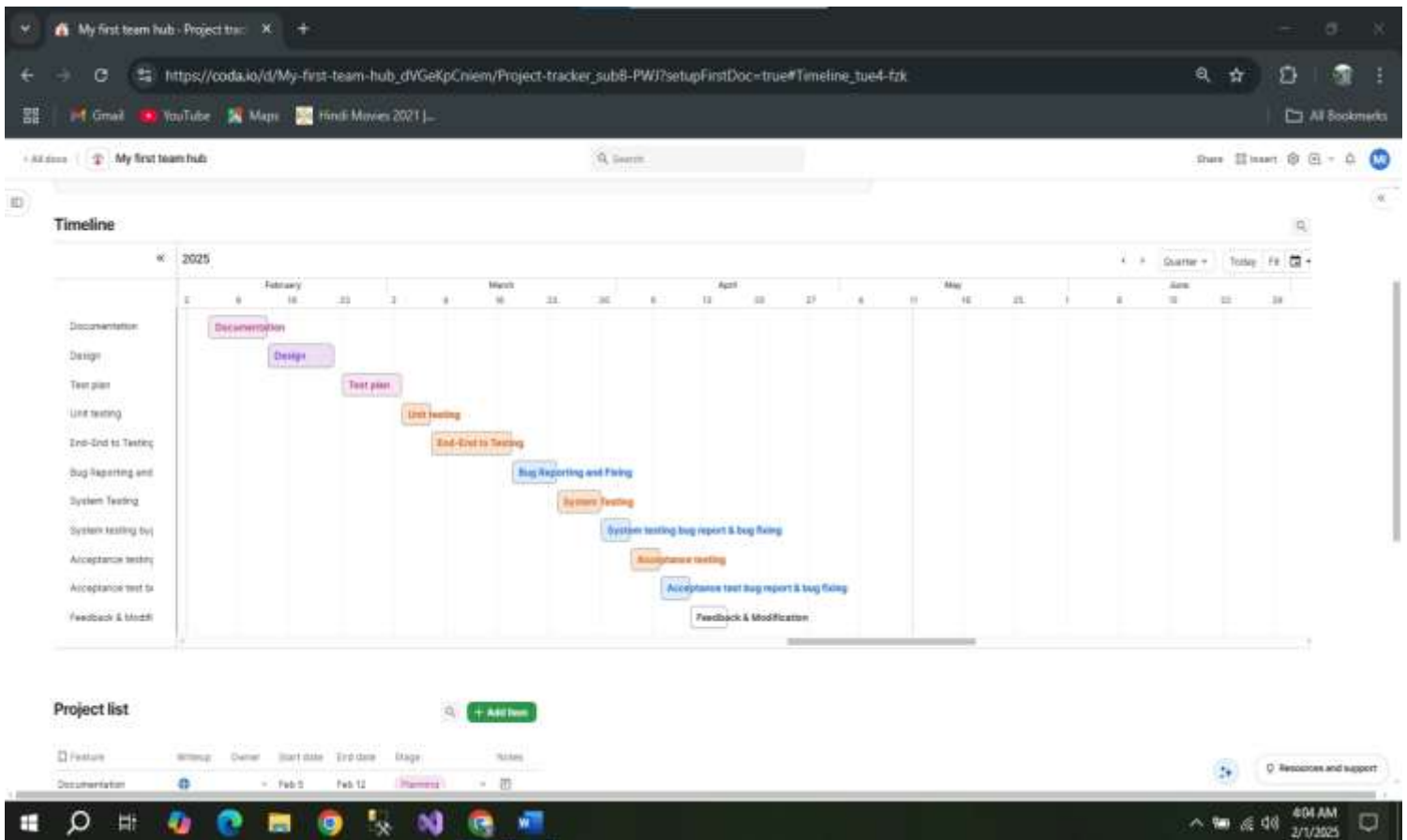
Training Needs:

- **Developers and Testers:** Training on core functionalities, third-party integrations, and platform operations.
- **Customer Support/Operations Staff:** Training on property management, booking issues, and user query resolution.
- **Property Owners/Tenants:** Tutorials on platform features like booking, messaging, and notifications.
- **Sales/Admin Staff:** Training on dashboards, reports, and analytics for tracking activity and revenue.

11. RESPONSIBILITIES

	TM	PM	Dev Team	Test Team	Client
Acceptance test Documentation & Execution					X
System/Integration test Documentation & Exec.				X	
Unit test documentation & execution			X		
System Design Reviews	X				
Detail Design Reviews		X			X
Test procedures and rules				X	
Screen & Report prototype reviews					X

12. TESTING SCHEDULE



13. PLANNING RISKS AND CONTINGENCIES

Risk: Limited availability of key stakeholders (property owners, tenants, or administrative staff). Currently, the system relies on active participation from these users for reviewing documents and participating in the Acceptance Testing phase. Any unavailability or delays from their end could impact on the project timeline.

Contingency:

- Adjust the schedule for document reviews and Acceptance Testing to accommodate stakeholders' availability.
- Ensure no part of the review or testing processes is bypassed.
- Employ alternative measures, such as virtual sessions or detailed documentation, to gather feedback efficiently.
- Assign backup personnel or additional testers to maintain progress despite stakeholder delays.

APPROVALS

Role	Name
Project Sponsor	AIUB
QA Lead	Mahfuz Ul Alom Badhan
Project Manager	Mahfuz Ul Alom Badhan
Business Analyst	MD. SHAHRIAR IBNE AMIN SHAKIL
Test Manager	MD. SHAHRIAR IBNE AMIN SHAKIL
Development Manager	MD MINHAJUL ISLAM